

Gael Berlanga Boemare

Contact: contact@gaelberlan.ga

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Passionate interest in mathematics, applied physics, statistics, and emerging technologies.

Education

Integrated Master of Engineering

The University of Cambridge

2022 – 2026

- **Courses:** Mechanical Engineering, Structures and Materials, Electrical and Information Engineering, Mathematical Methods.
- **Grades:** Upper Second-Class Honours, completed first year.

A levels

The British School of Brussels

2020 – 2022

- **Courses:** Mathematics, Physics, Further Mathematics - Decision 1 and Mechanics 1 modules.
- **Grades:** A*, A*, A*.

Experience

Engineering Intern

AGRO – OLEUM INGENIERÍA S.L.

Jul 2023 – August 2023

- Collaborated with a team of engineers to produce and design the new factory of San Miguel SA.
- Specialised in the drawing/editing of the factory plans using AutoCAD, using data made from 3D modelling simulation software.
- Presented plans, led and communicated to onsite builders.

Data Science Intern

Desbrest Institute of Epidemiology and Public Health

December 2023 – January 2024

- Developed algorithms for automated pulmonary health analysis from exhaled gases.
- Carried out a study on the effects of coastal weather conditions on strokes.

Projects

Development of Quantitative Trading Bot

Jul 2023 – August 2023

- Completed QuantConnect 2023 Python Algorithmic Trading A-Z course.
- Produced a trading strategy using a sequential model neural network in Python.

Mars Lander Simulation

Jul 2023 – August 2023

- Implemented orbital dynamics and numerical integrators in C++ to simulate a probe's orbit around Mars.
- Developed a control system to automatically land the probe from orbit.

Awards

Belgian Physics Olympiad 3rd place.

May 2021

- Placed third solving 5 physics brainteasers.

European Statistics Competition 1st place.

Oct 2020

- Collaborated with two other teammates to produce a 2-minute video, placing first out of almost 5000 teams.

Additional Information

Design: AutoCAD, SOLIDWORKS.

Programming: C++, Python, R.

Languages: French and Spanish, native/bilingual proficiency. English, full professional proficiency.

Hobbies: College Rugby, University Athletics, Recreational Programming.